Institute of Archaeology
Czech Acad Sci, Brno

ATRIUM

Transnational Access Scheme Computer Vision in Archaeology Training School, Brno 2026

Funded by the Atrium Project, and hosted by the Institute of Archaeology Czech Academy of Science, Brno

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Figure 1. Group photo of participants. *David Spáčil*

Opening Remarks:

Before I begin discussing the workshop, I would like to express my gratitude to the Atrium Project for funding such a great programme, and to the organisers at the Archaeological Institute for hosting me and facilitating such a lovely trip. My time in Brno was fabulous, and I learned so many new things that I look forward to sharing with my colleges. Overall, it was a great experience.

The Computer Vision in Archaeology Training School was a week long training course on the use of different applications of Machine Learning and Computer Vision on a variety of archaeological applications, from carved stone, to coins, to aerial images. We also had the chance to have a city tour of Brno and get to know one another.

DAY 1:

Day one started with some lovely introductions to our hosts and each other. Once everything was squared away, we jumped right into discussing the week ahead. We stopped for a lovely lunch at local places near the institute and jumped right into the practical bits of information. The afternoon started with a lecture on Agents, which are able to help our coding and task based work. Then, we went onto a session overviewing Python! In the evening, we had a lovely introduction and get-to-know each other in the lovely Institute break room with a selection of wonderful vegan treats, and delicious beer and wine.

Figure 2. Evening get-together at the Institute, Day 1. *David Spáčil.*

DAY 2:

Tuesday morning started off fresh with a lovely kick off into our practical session on artefact image datasets and data annotation. This session was a personal favourite of mine! We worked with a tool called CVAT which seemed a brilliant addition to my tool kit. Any ML archaeologist knows, data annotation is the most time consuming and involved process (at least I certainly find it so!). It essentially automatically removes backgrounds and labels outputs much quicker than a manual process. It uses the tool called Segment Anything. Then we moved onto a very fun game developed by ARUP, called Archaeotag. I must say, what a fabulous way to get the public involved and make the outsourcing of object labelling fun and free!

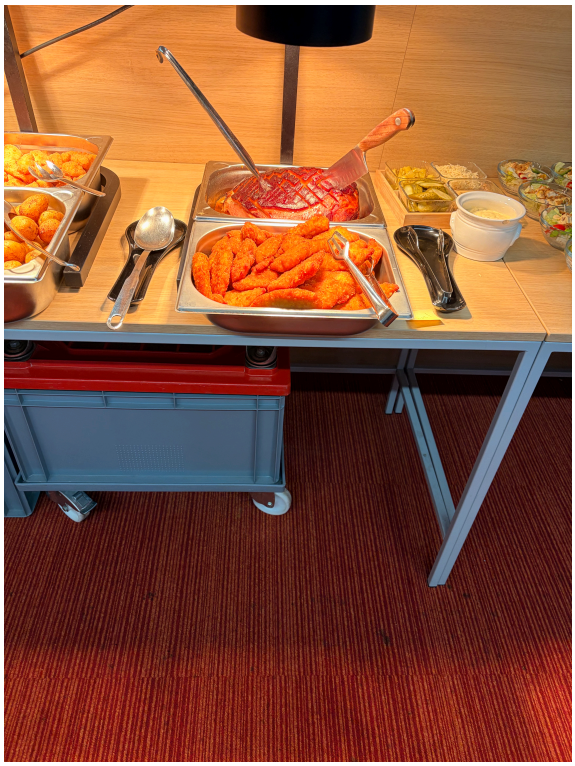
In the afternoon, we moved onto presentations from my fellow cohorts. It was lovely to hear the wide variety of different projects people were working on. And then, we took a lovely city tour of Brno, guided by Eva. Brno is a gorgeous, historical city, and for my first time in the Czech Republic, it left a great impression. We stopped at the end of the tour at a local pub, and I tried a ‘milky beer’ or a beer that essentially has a large foamy head on it. I was skeptical, but it was awesome! Czech beers are definitely famous for a reason.

Figure 3. Walking tour with Eva, always look up! *David Spáčil*

Day 3:

Wednesday started out just as nice, with a practical session on Microscopic images. It was very interactive and engaging, and I even got a chance to ask about Grad-CAM maps, a favourite tool of explainable data analysis of mine. I even got some great information on the 'edge-effect' and a source discussing the effect of objects of interest not being in the centre of the image not resulting in as great of accuracy- interesting!

The evening kicked off with a social, and wow, do the Czech's know how to host! Our lovely hosts at ARUP threw a mini party at the Institute. We were treated to an absolute feast! Ham, chicken, vegan and veggie treats, and a wide array of desserts. Also, I sampled some of the local wine, as Brno is quite close to a region well known for wine production. It was great! It was also lovely to get to know everyone a bit better.



Figures 4 & 5. Some of the special treats from our Wednesday evening social!

Day 4:

Day four focused on two different practicals. The morning, we looked at a series of Satellite imagery, probably the largest employed area of ML and Computer Vision in archaeology, so especially insightful and great to learn more about. In the afternoon, we moved onto rock art imagery- a personal favourite of mine! This session used a datasets from Scandinavia, which has an especially rich material culture and iconographic history of rock art. While we were waiting for

our models to finish processing, we got a tour of the Institute's library, which I got the chance to study at earlier in the day for my lunch break, as I went away just before my MPhil thesis was due. It was a great place to work, and a fabulous library. The Institute's facilities were really cutting edge.

Figure 6. The reading room of the library. *David Spáčil*

In the evening, a few of us joined up and went out for some traditional Czech food. It was delicious. We also ended up in a very random, but fun, dance party in the main Namesti (or Square). It was a great time out and lovely to further get to know some of the other participants.

Day 5:

Our final day was a bittersweet occasion, where our last session was on Ronny's coin dataset, and dilemma. It was all about how you must really think outside the box for archaeological datasets to find tools that will suit our needs as archaeologists. It rung true with my own research, and the solution to test whether coins are struck from the same die was to look to the sky, and used a matching tool for constellations! So inventive! As with any Computer Vision task, trial and error is often the way things go.

We finished up with a sentimental goodbye to the lovely friends we made throughout the week and said a final thank you to our gracious hosts. In the evening, I settled down in my hotel, and reflected on the wonderful week. I spent my final night on the roof terrace of my hotel, overlooking Brno's gorgeous skyline. What a way to end the trip!

Figure 7. Brno city skyline at night. A great end to my trip!

Final Reflections:

All in all, I had a wonderful time in Brno, and am so grateful that I got to take part in this amazing experience. I have learned so much, and met so many wonderful new friends. I do hope that I will be back to Brno in the future, and the week exceeded all my expectations. Thank you to everyone who made it possible!